

AWS Cloud Capacity Shortage

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Cloud computing has become the backbone of modern technology with how often organizations rely on remote storage, online resources, and on demand computing power for their everyday operations. It feels like everything runs in the cloud now from business applications to entertainment, banking, healthcare, and government systems. Amazon Web Services is one of the most widely used cloud providers in the world and because of how large and successful it is, it has recently become a victim of its own growth. AWS has been facing ongoing capacity shortages due to overwhelming demand for storage, computing power, artificial intelligence training, and general cloud resources that continue to increase every year. This situation became even more complicated because AI technologies develop extremely fast, meaning AWS has to constantly scale its infrastructure just to keep up. The shortages reported in the last twelve months have impacted companies globally and have affected regular users, large corporations, and even certain government operations.

The event I selected covers the recent reporting from iTWire about AWS struggling to meet customer demand because of data center capacity issues. According to iTWire, AWS does not have enough available computing power in some of its regions which has forced customers to delay their projects, pause deployments, and in some situations completely rethink their business plans. The shortages resulted in AWS restricting access to certain high demand cloud resources normally used for artificial intelligence and large data processing. These restrictions caused organizations to wait weeks or months for resources they normally could access instantly. Since AWS is trusted by so many businesses as their main cloud provider, these shortages reveal how risky it can be for the tech world to rely too heavily on a single provider.

The importance of this event is tied directly to information and technology policy because cloud computing is now treated as critical infrastructure. When a provider like AWS

struggles to keep up with demand, the effects can be felt across entire industries. It raises questions about whether companies should rely on one central cloud service for everything. It also raises questions about whether the government should introduce policies that require cloud providers to meet certain capacity, stability, and resiliency standards, especially when services impact financial institutions, hospitals, government websites, public safety systems, and national security programs. These shortages also highlight how laws and regulations barely keep up with how fast cloud services grow.

When reviewing congress.gov for laws that address this specific problem, there is no single law that directly deals with cloud capacity shortages or cloud provider resource limits. To confirm this, I searched for several specific phrases including cloud capacity shortage, cloud infrastructure shortage, AWS capacity, cloud computing limitations, cloud provider regulation, and commercial data center reliability. None of these searches produced legislation focused on the type of capacity problems AWS is experiencing. The closest related results involve general cybersecurity laws, federal cloud adoption policies, and national critical infrastructure protection programs, but none address cloud resource shortages. Based on these searches, I believe there is currently no law that directly deals with this event. Even though cloud computing is a very important part of today's infrastructure, policy makers have not yet created regulations that address supply limits or availability standards for commercial cloud services.

While there may not be a direct law for this issue, a few existing federal policies relate indirectly. For example, the Federal Risk and Authorization Management Program (also known as FedRAMP) sets standards for cloud providers working with government agencies. However, FedRAMP focuses on security, compliance, and authorization and does not address cloud capacity, allocation, or shortages. This confirms that the AWS capacity problem is something the

current legal system does not fully address. If cloud computing continues to expand at this rate, lawmakers might eventually need to consider new legislation to ensure cloud providers can maintain availability and stability at a level that matches their importance in society.

Several stakeholders are impacted by this event including businesses, government agencies, software developers, everyday cloud users, global customers, and AWS itself. Businesses that rely on the cloud for day to day operations are impacted the most because they need consistent, reliable access to computing power to run their applications. When AWS cannot provide resources, these businesses experience delays, downtime, or have to spend more money to rework their plans. Government agencies are also affected because many public systems use AWS for authentication, data storage, and communication services. Software developers and IT professionals are impacted because their projects cannot move forward when resources are restricted. Everyday users who depend on websites and apps hosted in AWS regions may experience outages or slower performance. Global customers are heavily affected because international projects depend on AWS capacity in their regions. AWS is also a stakeholder because these shortages harm its reputation and financial stability.

There are clear ethical outcomes related to this event when using Quinns workable ethical theories. Using utilitarian theory, AWS has a responsibility to act in a way that benefits the largest amount of people. In this case, failing to provide enough resources negatively affects thousands of customers and millions of users. From a duty based ethics perspective, AWS has a duty to provide reliable services to customers who trust them with critical workloads. If AWS knowingly sells more services than it can support, this becomes an ethical concern. Another ethical concern is transparency. Some customers may not have been aware of the shortages until after problems occurred. A lack of transparency can be seen as deceptive if customers pay for

resources that are not available. On the other hand, from a fairness perspective, AWS must distribute limited resources in a fair way which can be difficult when demand is extremely high. Overall, this event does involve ethics because it deals with responsibility, fairness, and the level of trust customers place in cloud providers.

This event also raises concerns for future information technology policy. Policymakers might need to consider whether cloud providers should be treated more like public utilities. If the entire country relies on cloud computing for banking, healthcare, government services, and communication, then outages or shortages are similar to losing electricity or water. There might be a need for federal standards about maintaining minimum levels of capacity and requiring providers to disclose shortages ahead of time. Another possible policy approach could involve requiring more competition in the cloud market so customers do not depend on one company too heavily. As new forms of artificial intelligence continue to expand, demand for cloud hardware will only increase. Without strong policy guidance, shortages like this could repeat or get worse.

In conclusion, researching this event has helped me understand how complex and important cloud computing has become. Before this, I knew cloud services were important, but I did not realize how much pressure companies like AWS face when trying to scale fast enough for global demand. It was interesting to see how an issue that looks purely technical actually has major policy implications and ethical consequences. This process also showed me how far behind legislation can be compared to technology. Even though AWS is widely used for critical operations, there is still no law that deals with cloud capacity shortages or resource availability. Overall, this event demonstrates how important it is for policymakers, cloud providers, and customers to work together to make sure cloud services remain reliable, transparent, and stable for the future.

References

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